Machine Vision Tool Tracking System

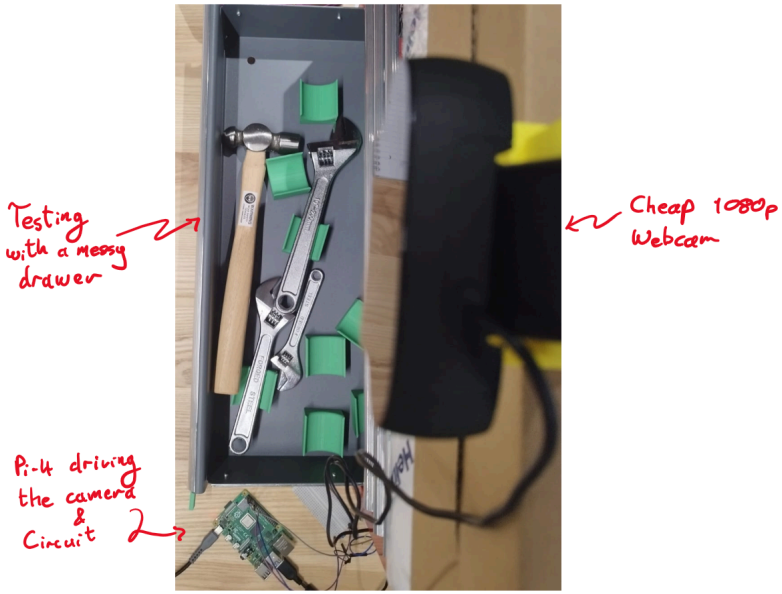
The School of Engineering at Trinity College Dublin hosted a hackathon for a few of the engineering societies as part of the official opening of the E3 Learning Foundry. I took part as a member of the Trinity Floating Wind team.
They gave us a brief that centered around the problem of how students were supposed to share tools in the new makerspace in a way that ensured accountability and proper management of tools. There were 6 toolboxes already in place so we had to build a system on top of that.

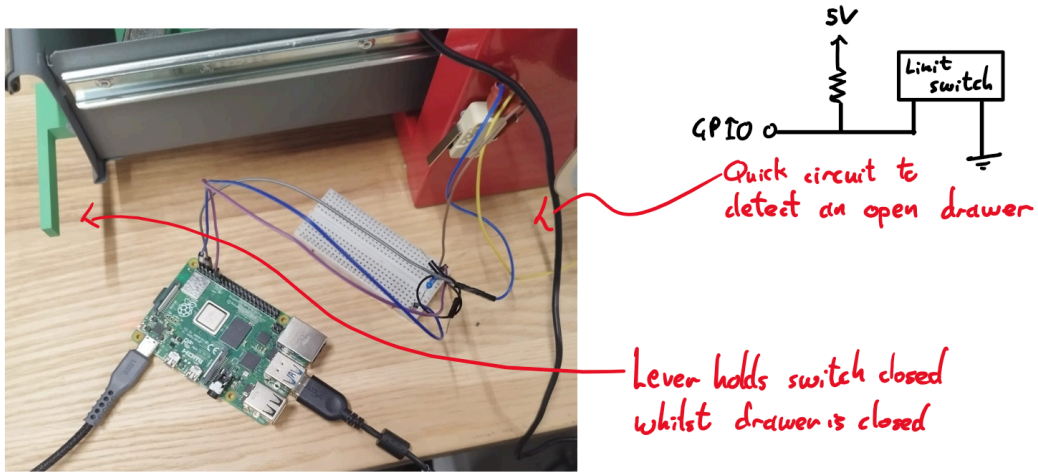


The tool box

Our solution centred around machine vision. A raspberry pi would detect when a drawer was opened, take photos, send them to an app which interacts with a cloud-hosted version of MobileNetV2 which we fine-tuned with specific images and made note of whether a tool was present in the drawer or not.

https://github.com/Adam76e54/machine-visions-tts



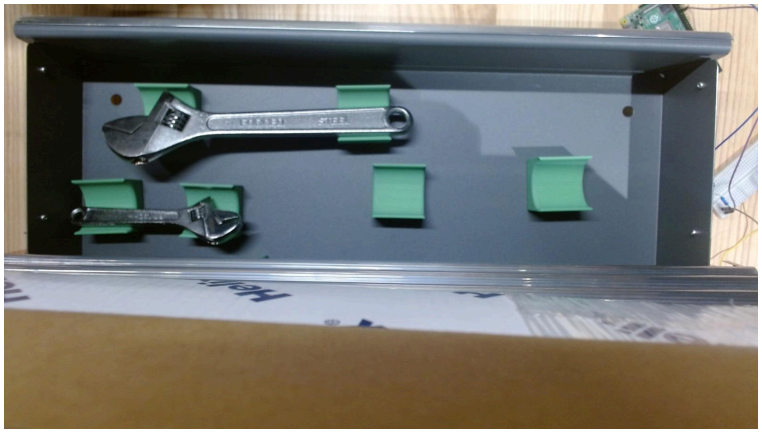


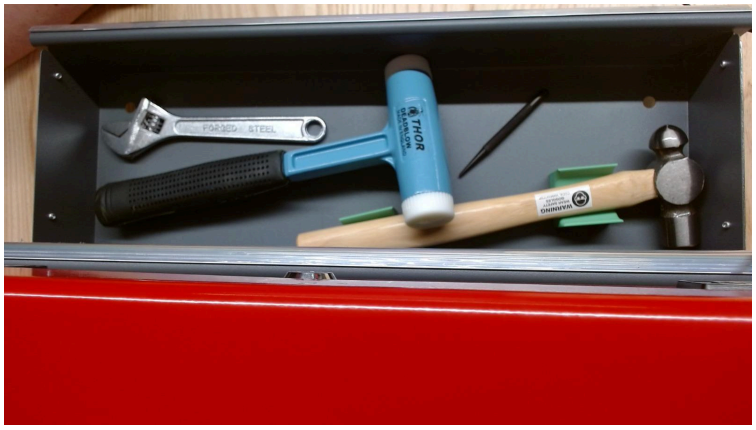
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Some notes:

- The switch-based sensor is quite scrappy. A better solution would be to have an infrared sensor at the back of the drawer that gathers real-time data about how far open the drawer is & whether it is opening or closing.
- The network was most confident in scenarios where tools are but is still robust enough to be correct on slightly messy drawers.
- Around 130 images were taken with the webcam to fine-tune Google's MobileNetV2 using pytorch.
- A UI was generated using Claude Code CLI to demonstrate the prototype.
- The camera stand, tool-holders, and switch lever were 3D printed.

There is a github showing some more images & the code (still in the process of being added-to as of 29/10/2026)



Examples of training images used

https://github.com/Adam76e54/machine-visions-tts